

Maven AI Investment Copilot – Engineering Decisions, Trade-offs & Future Improvements

1. Engineering Decisions

- **Hybrid AI Architecture:** Financial calculations are deterministic while LLMs are reserved for reasoning, interpretation, and explanation. This improves reproducibility and reduces hallucinations.
- **Evidence-First Pipeline:** Every recommendation is derived from structured evidence before any AI synthesis occurs, improving transparency.
- **AI Investment Committee:** A dedicated committee stage reviews the investment thesis independently before a recommendation is issued, encouraging balanced reasoning.
- **Self-Critique Layer:** A critique stage challenges assumptions, highlights missing evidence, and increases trust in the final recommendation.
- **Microservice Design:** The React frontend, Node.js gateway, and Python AI service are separated to improve maintainability, scalability, and independent deployment.
- **Report-Level Caching:** Completed reports are cached to avoid repeated AI calls, reducing latency and API costs while preserving consistency.

2. Key Trade-offs

- **Custom Orchestration vs LangChain:** A custom orchestration layer was chosen to keep the MVP lightweight and fully understood. It offers fine-grained control but lacks the ecosystem benefits of LangChain/LangGraph.
- **Single LLM Provider:** Google Gemini was selected for simplicity. Supporting multiple providers was intentionally deferred.
- **Single Company Analysis:** The MVP analyzes one company at a time. Portfolio analysis and comparisons were postponed to keep the investment pipeline focused.
- **Live Data over Historical Knowledge:** The system prioritizes current public financial information instead of maintaining a large internal financial knowledge base.
- **Explainability over Complexity:** The architecture favors transparency and deterministic reasoning instead of maximizing autonomous AI behavior.

3. Features Intentionally Left Out

- Portfolio optimization and watchlists
- Historical backtesting
- Real-time monitoring and alerts
- Company-to-company comparison
- User authentication and collaboration
- PDF export and scheduled research
- Autonomous specialist agent teams

4. Future Improvements

Phase 1 – LangChain / LangGraph

Replace the custom orchestration layer with LangChain/LangGraph to gain workflow state management, reusable chains, observability, and easier integration of specialized reasoning agents while preserving deterministic financial analysis.

Phase 2 – Retrieval-Augmented Generation (RAG)

Introduce Retrieval-Augmented Generation using a vector database to retrieve relevant evidence from SEC filings, annual reports, quarterly reports, earnings call transcripts, investor presentations, previous research reports, and trusted news before reasoning.

Phase 3 – Vector Database

Integrate a vector database such as Pinecone, Qdrant, Weaviate, or Chroma to support semantic search, long-term report memory, retrieval of historical analyses, and evidence-backed follow-up conversations.

Phase 4 – Advanced AI Capabilities

- Specialized agents for valuation, governance, business quality, and risk analysis.
- Portfolio-level investment intelligence.
- Continuous monitoring of tracked companies.
- Interactive explanation and challenge workflows.
- Evidence citations linked directly to supporting documents.
- Multi-model support with configurable LLM providers.

5. Conclusion

Maven was intentionally designed as an explainable AI investment copilot rather than a generic chatbot. The MVP prioritizes transparency, modularity, and deterministic reasoning. Future work focuses on richer retrieval, workflow orchestration with LangChain/LangGraph, semantic memory, and enterprise-scale AI capabilities.